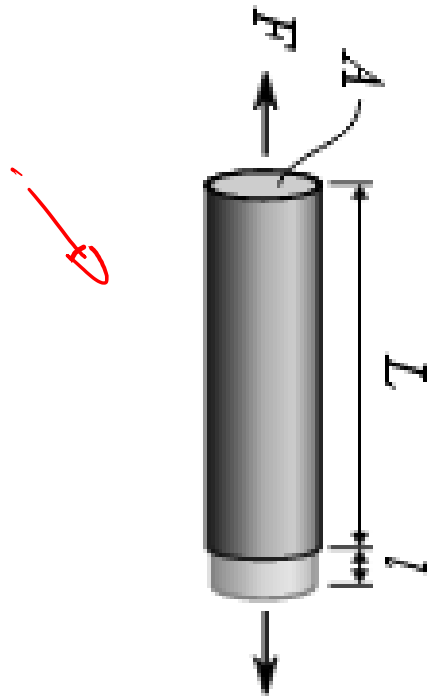


A

STM Exercise 2



$$\sigma = \frac{F}{A}$$

where

- σ stress [Pa]
- F applied force [N]
- A cross-sectional area [m²]

$$\varepsilon = \frac{\Delta L}{L_0}$$

where

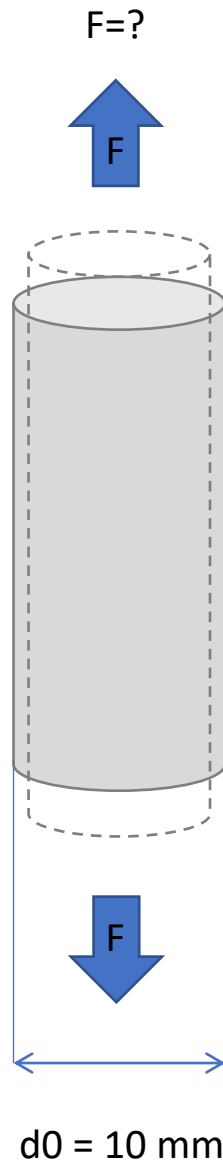
- ε strain
- ΔL total elongation [m]
- L_0 original length [m]

$$E = \frac{\sigma}{\varepsilon}$$

Exercise 1: You have a cylindrical component made of an Al alloy (10 mm diameter, YS= 150 MPa): which is the maximum load you can apply on this component without having plastic deformation? Consider a safety factor of $n = 2$.

σ_y

1-You have a cylindrical component made of an Al alloy (10 mm diameter, $Y_S = 150 \text{ MPa}$): which is the maximum load you can apply on this component without having plastic deformation? Consider a safety factor of $n = 2$.

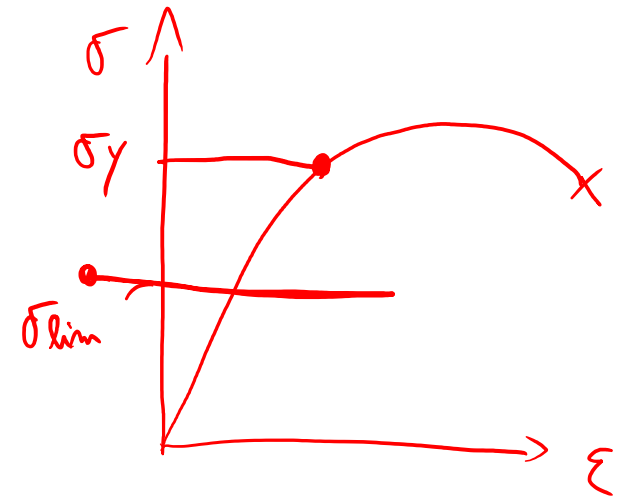


$$\sigma_y \quad \sigma_{lim} = \frac{\sigma_y}{2} = \frac{150 \text{ MPa}}{2} = 75 \text{ MPa}$$

$$\sigma = \frac{F}{A} \quad A = \pi \cdot r^2 = 78,5 \text{ mm}^2$$

$$F = \sigma \cdot A = 75 \text{ MPa} \cdot 78,5 \text{ mm}^2 = 5887,5 \text{ N}$$

$$\left[\frac{\text{N}}{\text{mm}^2} \right] \cdot \text{mm}^2$$



Poisson's ratio, ν

• Poisson's ratio, ν :

$$\nu = \frac{-\epsilon_x}{\epsilon_z} = \frac{-\epsilon_y}{\epsilon_z}$$

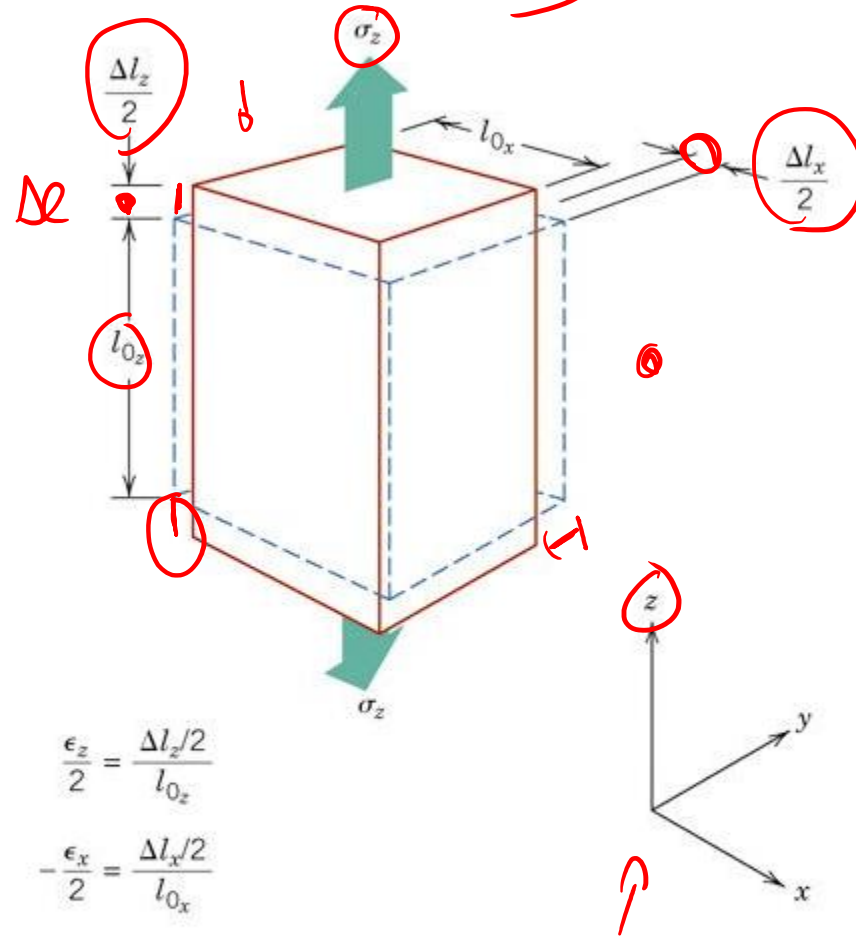
(Isotropic materials only)

metals: $\nu \sim 0.33$
ceramics: $\nu \sim 0.25$
polymers: $\nu \sim 0.40$

Units:

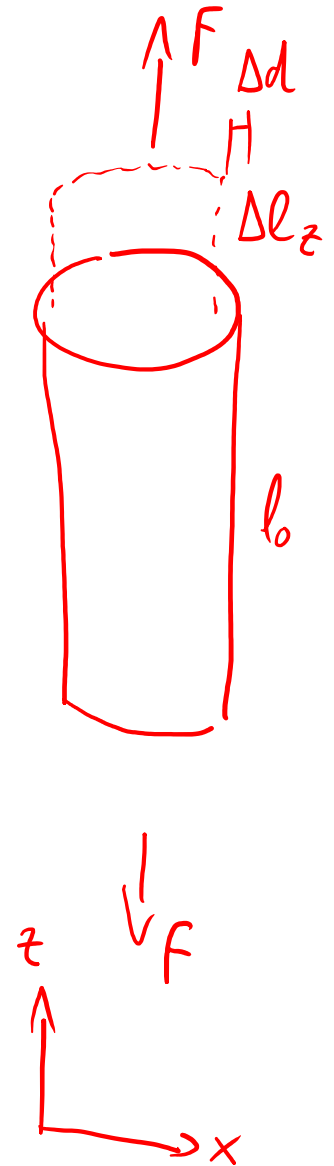
E : [GPa] or [psi]

ν : dimensionless



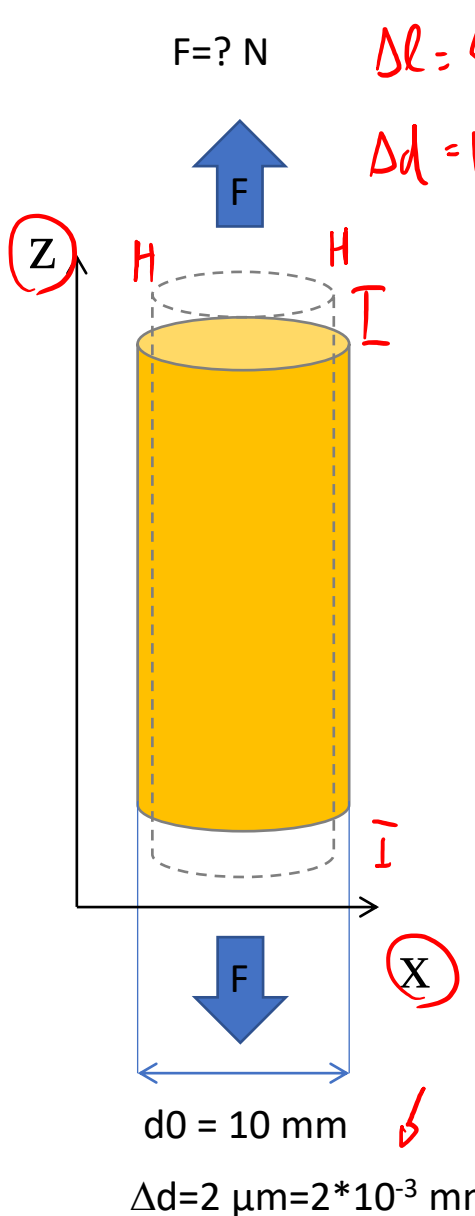
$$\frac{\epsilon_z}{2} = \frac{\Delta l_z / 2}{l_{0z}}$$

$$-\frac{\epsilon_x}{2} = \frac{\Delta l_x / 2}{l_{0x}}$$



Exercise 2: You have a cylindrical aluminum component (diameter =10 mm, $\nu=0.34$, $E=70$ GPa): calculate the tensile load you need to apply to have a 2 micron smaller diameter component.

2- You have a cylindrical aluminum component (diameter = 10 mm, $\nu=0.34$, $E=70 \text{ GPa}$): calculate the tensile load you need to apply to have a 2 micron smaller diameter component.



Poisson (ν) = $-\epsilon_x / \epsilon_z$

$$\epsilon = \frac{\Delta L}{L_0} = \frac{\Delta d}{d_0}$$

When a load F is applied the component elongates along z and reduce its diameter along x .

$$\epsilon_x = \frac{\Delta d}{d_0} = \frac{-2 \times 10^{-3} \text{ mm}}{10 \text{ mm}} = -2 \times 10^{-4}$$

(negative deformation for diameter reduction)

$$\epsilon_z = -\frac{\epsilon_x}{\nu} = -\frac{(-2 \times 10^{-4})}{0.34} = 5.8 \times 10^{-4}$$

$$\sigma = \epsilon_z E = (5.8 \times 10^{-4}) (70 \times 10^3 \text{ MPa}) = 40.6 \text{ MPa}$$

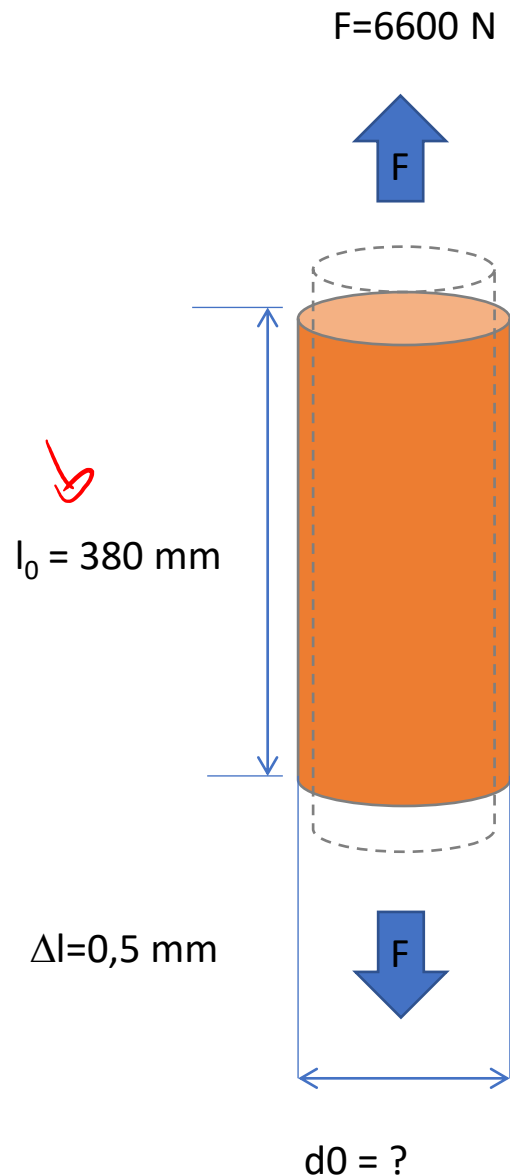
$$A = \pi r^2 = 78.5 \text{ mm}^2$$

$$F = \sigma * A = 40.6 * 78.5 = 3187 \text{ N}$$

$$\sigma = \frac{F}{A}$$

Exercise 3: Which is the diameter you need for a 380 mm long Cu component ($E=110$ GPa) to be loaded at 6600N, if you can accept an elongation of 0.5 mm.

3- Which is the diameter you need for a 380 mm long Cu component ($E=110$ GPa) to be loaded at 6600N, if you can accept an elongation of 0.5 mm. Δl



$$\varepsilon = \Delta l / l_0 = 0.5 / 380 = 0.0013$$

$$\rightarrow \sigma = E * \varepsilon = 110000 * 0.0013 = 144.7 \text{ MPa}$$

$$\rightarrow \sigma = F / A$$

$$A = F / \sigma = 6600 / 144.7 = 45.6 \text{ mm}^2$$

$$A = \pi * r^2$$

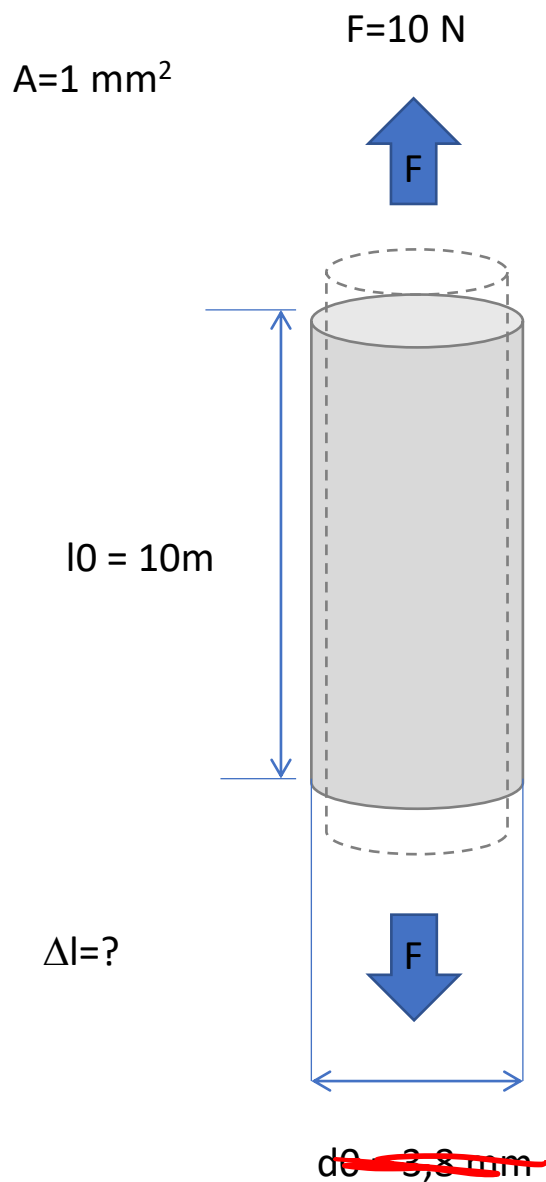
$$\rightarrow R = \sqrt{\frac{A}{\pi}} = 3.8 \text{ mm}$$

$$D = 2 * r = 7.6 \text{ mm}$$

$$\frac{[N]}{[N/mm^2]}$$

Exercise 4: The elastic modulus of a steel is 193 GPa, of a bronze is 100 GPa, and of a polymer is 0,17 GPa ; calculate the elastic elongation of a 10 meter long wire made of steel, bronze or polymer , cross section is 1 mm², the applied load is F=10 N.

The elastic modulus of a steel is 193 GPa, of a bronze is 100 GPa, and of a polymer is 0,17 GPa ; calculate the elastic elongation of a 10 meter long wire made of steel, bronze or polymer , cross section is 1 mm², the applied load is F= 10 N.



$$\varepsilon = \Delta l / l_0$$

$$\Delta l = \varepsilon * l_0$$

$$\varepsilon = \sigma / E$$

$$\sigma = F / A = 10 / 1 = 10 \text{ MPa}$$

It is the same for all the materials

$\varepsilon = \sigma / E$ depends on the Elastic Modulus (E), different for each material

$$\varepsilon_{\text{steel}} = 10 / 193000 = 5.18 * 10^{-5} \rightarrow \Delta l_{\text{steel}} = 5.18 * 10^{-5} * 10000 = 0.518 \text{ mm}$$

$$\varepsilon_{\text{bronze}} = 10 / 100000 = 0.0001 \rightarrow \Delta l_{\text{bronze}} = 0.0001 * 10000 = 1 \text{ mm}$$

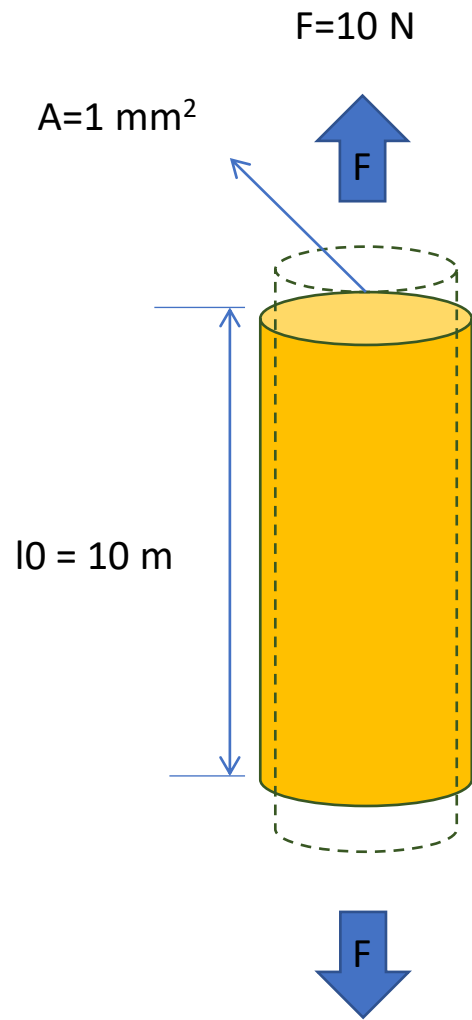
$$\varepsilon_{\text{polymer}} = 10 / 170 = 0.06 \rightarrow \Delta l_{\text{polymer}} = 0.06 * 10000 = 600 \text{ mm}$$

0,6 m

$$l_0 = 10 \text{ m} = 10000 \text{ mm}$$

$$A = 1 \text{ mm}^2$$

Exercise 5: A wire 10 m long with a cross section of 1mm^2 underwent tensile loading of 10N. Determine stress, strain and elongation if the wire is made of Copper ($E=110\text{ GPa}$) or steel ($E=210\text{ GPa}$). Compare different material on the available excel file.



$$\sigma = F/A = 10 \text{ [N]}/1 \text{ [mm}^2\text{]} = 10 \text{ MPa}$$

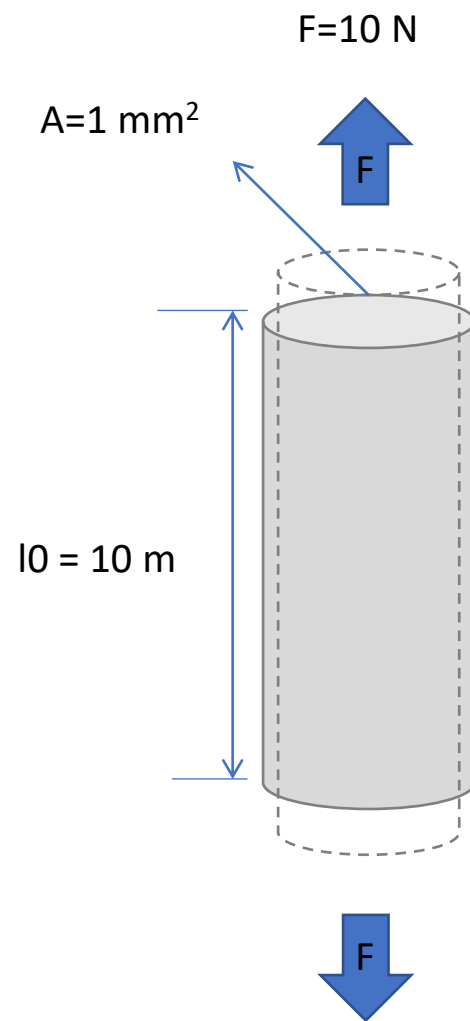
If the wire is made of **copper** ($E = 110 \text{ GPa}$) the elastic deformation is:

$$\sigma = E * \varepsilon \rightarrow \varepsilon = \sigma / E$$

$$\varepsilon = 10 \text{ [MPa]}/110\,000 \text{ [MPa]} = 0.00009 = 0.009 \%$$

$$\varepsilon = \Delta l / l_0 \quad \Delta l = \varepsilon * l_0$$

The total elongation on 10 m is
 $0.00009 \times 10 \text{ m} = 0.0009 \text{ m} = 0.9 \text{ mm}$



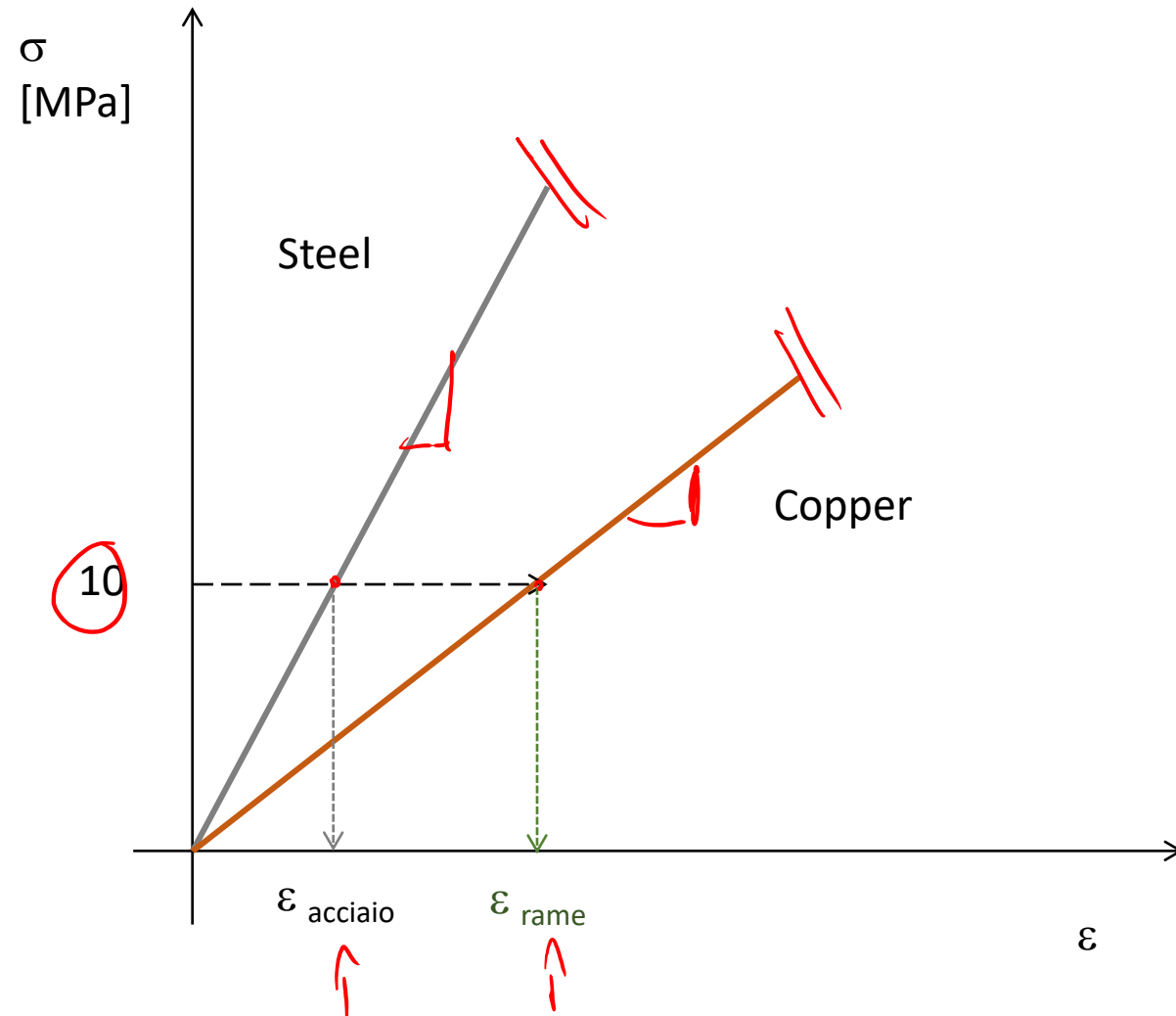
If the wire is made of **steel** ($E=210 \text{ GPa}$):

$$\varepsilon = 10 / 210\,000 = 0.000048$$

The total elongation on 10 m is

$$0.000048 \times \underline{10 \text{ m}} = 0.00048 \text{ m} = \underline{\underline{0.48 \text{ mm}}}$$

Since the elastic modulus of steel is almost double of the one of copper, the elastic deformation, in the same conditions, is almost a half.



Exercise 6



Consider an aluminium cylindrical sample with a diameter of 1 cm.
the significant values for applied force underlined in tensile test are:

- 4600 N (yield),
- 8450 N (maximum)
- 5700 N (rupture) .

When rupture occurs sample diameter is $\frac{1}{3}$ of the initial one

Determine:

- Yield strength
- Tensile strength
- Seemly fracture strength
- True fracture strength

Initial section: $(5)^2 \cdot \pi \text{ [mm}^2\text{]} = 78.5 \text{ mm}^2$.

$R = 5 \text{ mm}$

Yield strength:

$$\sigma_y = 4600 / 78.5 = 58.6 \text{ N/mm}^2 = 58.6 \text{ MPa}$$

Tensile strength:

$$\sigma_m = 8450 / 78.5 = 107.6 \text{ N/mm}^2 = 107.6 \text{ MPa}$$

Seemly fracture strength:

$$\sigma_R = 5700 / 78.5 = 72.6 \text{ N/mm}^2 = 72.6 \text{ MPa}$$

A_0

$$R = 1,65 \text{ mm}$$

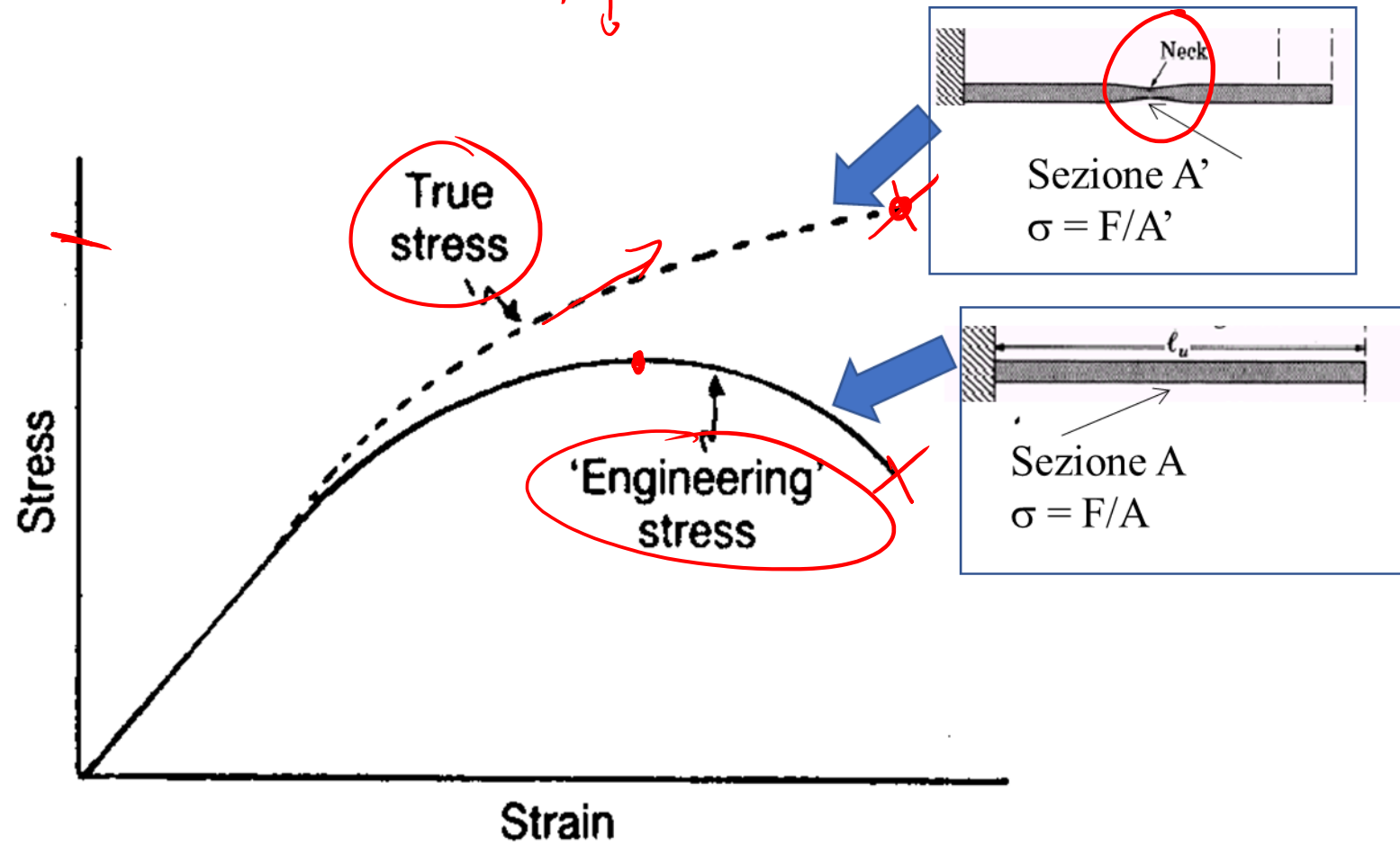
If at rupture the sample has a diameter of 1/3 of the initial one, so 0.33 cm (3.3 mm), the real section supporting the load is only:

$$A = (1.65)^2 \cdot \pi [\text{mm}^2] \Rightarrow 8.6 \text{ mm}^2$$

So the **true fracture strength** is:

$$\sigma_R = \underline{5700} / \underline{8.6} = 662.8 \text{ N/mm}^2 = \mathbf{662.8 \text{ MPa}}$$

$$\sigma = \frac{F}{A}$$



A_0